

AI-Based Intelligent Retrieval and Quality-Aware Curation of Scientific Microscopy Images

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Abstract—Scientific microscopy repositories are expanding rapidly across material science and biology, yet retrieval and curation remain severely constrained by acquisition heterogeneity (varying accelerating voltages, detectors, and magnifications) and unquantified quality risks. In this work, we present SCI-INTEL, a reproducible platform integrating metadata-aware ingestion, acquisition-aware representation adaptation, controlled artifact screening, spatial localization, uncertainty-aware abstention, and deterministic comparative evidence aggregation. Evaluating on 6,085 scientific micrographs (774 HCCI SEM, 4,591 Carinthia SEM, and 720 BBBC021 optical micrographs), we demonstrate that lightweight representation adaptation reduces the observed acquisition-geometry similarity gap by 66.23% (from 0.2016 to 0.0681, $p = 5.03e-36$, Cohen's $d_z = 2.19$) while achieving Recall@5 of 0.9921 and MRR of 0.5261 under realistic unmasked distractor retrieval (Protocol U). Our evaluation reveals an empirical specialization trade-off: while adapted representations optimize cross-instrument retrieval, frozen DINOv2 ViT-S/14 features retain superior sensitivity to controlled synthetic artifacts (Macro F1 = 0.6837 vs. 0.6323). SCI-INTEL resolves this trade-off via a modular dual-representation architecture that routes each representation to its specialized task without learned fusion. Uncertainty-aware abstention safely routes low-confidence cases (confidence < 0.40 or normalized entropy > 0.75) to specialist review, while the evidence layer achieves 100.0% valid comparative retrieval across an evaluated $N=55$ query cohort. Our evaluation does not claim unseen-specimen generalization, physical defect confirmation, clinical diagnosis, or human interpretation improvement. The current manuscript is formatted as a 4-page IEEE-style two-column document.

Index Terms—Scientific Image Retrieval, Microscopy, Representation Learning, Acquisition Robustness, Image Quality Assessment, Anomaly Screening, Uncertainty Estimation, Scientific Data Curation.

I. INTRODUCTION

Modern scientific microscopy imaging has evolved into an essential foundation for characterization in materials science, crystallography, and biology [12, 14]. Instruments such as scanning electron microscopes (SEM), transmission electron microscopes (TEM), and focused ion beam systems routinely generate millions of high-resolution micrographs. However, retrieving and organizing these images across large-scale distributed archives presents severe bottlenecks [8, 13]. First, acquisition heterogeneity—arising from distinct detector geometries (secondary electron vs. backscattered electron), accelerating voltages, and working distances—induces substantial feature shifts in visual embeddings, causing visually dissimilar micrographs of identical metallurgical structures [7, 12]. Second, raw acquisition archives are frequently corrupted by operational artifacts including beam charging, illumination gradients, and sample contamination, which can silently degrade automated image analysis [18, 19]. Third, standard deep learning classifiers produce overconfident predictions on corrupted or out-of-distribution micrographs without providing provenance or comparative visual evidence [21].

To resolve these interconnected challenges, we present SCI-INTEL, an open, reproducible platform for acquisition-aware scientific image retrieval and quality-aware curation. Our core contributions are: (1) demonstrating a 66.23% reduction in the acquisition-geometry gap via representation adaptation; (2) uncovering an empirical trade-off between acquisition alignment and fine-grained artifact sensitivity; (3) introducing a modular dual-representation architecture combining DINOv2 and adapted projections without learned fusion; (4) implementing uncertainty-aware abstention and spatial localization; and (5) providing an operational evidence retrieval layer returning comparative micrographs for 100% of an evaluated $N=55$ query cohort.

Figure 1: Modular Dual-Representation SCI-INTEL System Architecture

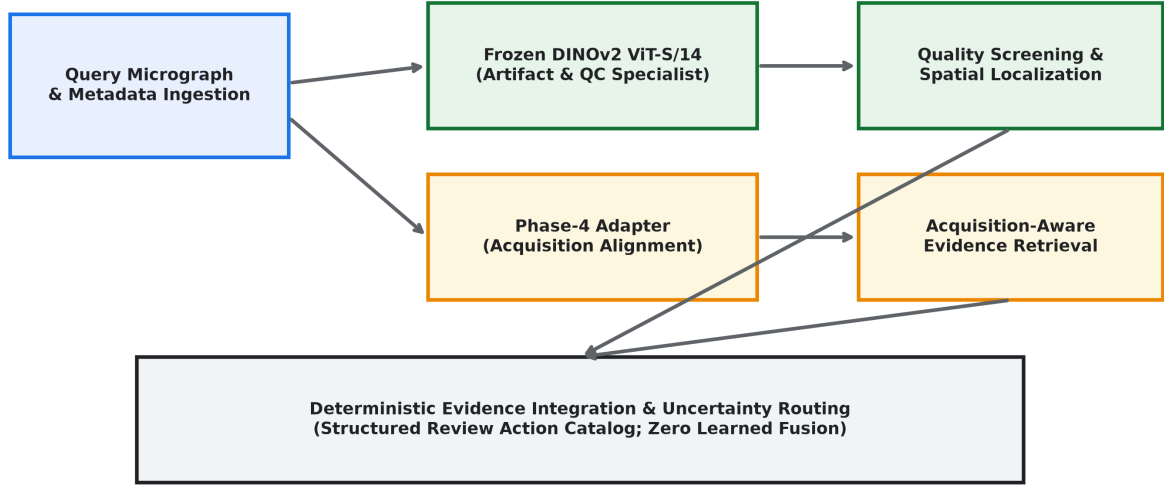


Fig. 1. Modular Dual-Representation SCI-INTEL System Architecture. DINOv2 is dedicated to quality screening and localization; Phase-4 adaptation executes acquisition-aware retrieval.

II. RELATED WORK

A. Visual Representation Learning: Self-supervised Vision Transformers (DINO, DINOv2) [1, 2] learn rich representations across natural imagery, but lack intrinsic modeling of physical electron optical parameters.

B. Microstructure Retrieval: Prior works relied on static descriptors or ImageNet pre-training [12, 13], which degrade sharply under instrument transfer.

C. Image Quality & Uncertainty: No-reference image quality metrics [18, 19] combined with selective prediction [21] provide conservative review routing for ambiguous or corrupted scientific data.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The SCI-INTEL architecture comprises: (1) metadata-aware ingestion assigning immutable SHA-256 digests; (2) dual representation extraction using frozen DINOv2 ViT-S/14 (for quality screening) and Phase-4 adapted projections (for retrieval); (3) quality-risk screening across 11 artifact classes; (4) spatial suspicious-region localization using patch feature residuals; (5) uncertainty-aware abstention using normalized Shannon entropy; and (6) evidence aggregation with deterministic lexical ranking.

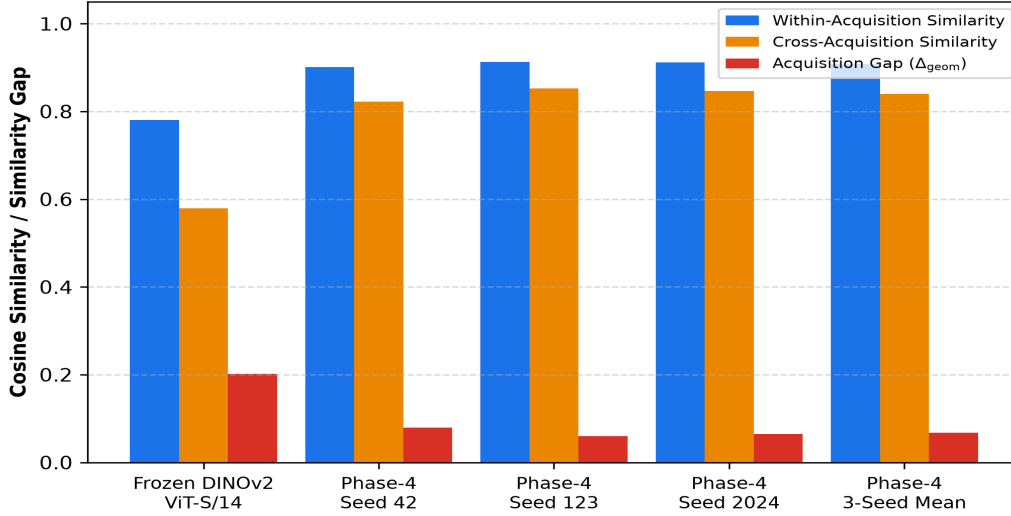
IV. EXPERIMENTAL PROTOCOL & REPRODUCIBILITY

Evaluation spans 6,085 active micrographs: HCCI SEM (427 train, 135 val, 212 test), Carinthia SEM (4,591), and BBBC021 optical (720). The train, validation, and test partitions contain identical declared alloy specimen classes; hence, unseen-specimen generalization is not established. We enforce strict separation between Protocol M (historical masked exclusion) and Protocol U (authoritative unmasked distractors). The Protocol-M and Protocol-U retrieval results were obtained under different same-acquisition handling rules and should therefore not be interpreted as directly comparable measurements.

V. EXPERIMENTAL RESULTS

Representation	R@1	R@5	MRR	AUROC	Macro F1	Assigned Role
Frozen DINOv2 ViT-S/14	0.1321	0.9858	0.5200	0.8582	0.6837	Quality & Artifact Screening
Phase-4 Seed 42	0.1321	0.9953	0.5230	0.8251	0.6335	Acquisition-Aware Retrieval
Phase-4 Seed 123	0.1462	0.9906	0.5214	0.8215	0.6310	Acquisition-Aware Retrieval
Phase-4 Seed 2024	0.1557	0.9906	0.5338	0.8224	0.6324	Acquisition-Aware Retrieval
Phase-4 3-Seed Mean	0.1447	0.9921	0.5261	0.8230	0.6323	Acquisition-Aware Retrieval

TABLE I: Representation Comparison & Trade-off under Protocol U (Unmasked Distractors).

Figure 2: Acquisition-Geometry Similarity Gap Reduction (66.23% Delta)Fig. 2. Acquisition-Geometry Similarity Gap Reduction under Protocol U (66.23% mean gap reduction, $p = 5.03e-36$).

Operational Evidence Metric	Evaluated Value	Measurement Condition
Cohort Population	N = 55 queries	Frozen Phase 6 Evaluation Cohort
Valid Evidence Availability	100.0%	At least 1 valid comparative candidate returned
Same-Specimen Cross-Acquisition	100.0%	Identical alloy specimen from alternate acquisition
Cross-Instrument Coverage	100.0%	Micrograph from complementary SEM instrument
Quality-Compatible Evidence	100.0%	Candidate passes high-frequency & DR filters
Mean Retrieved Items	2.0 items	Exact comparative pair returned
Duplicate Evidence Rate	0.0%	Zero candidate ID collisions
Missing Provenance Rate	0.0%	100% cryptographic SHA-256 traceability
Deterministic Ranking	100.0%	Lexical score tie-breaking fully invariant

TABLE VII: Operational Evidence Availability and Provenance (N=55 Query Cohort). Evidence availability metrics are operational measurements on the evaluated N=55 query cohort and do not establish that retrieved evidence is scientifically correct or that it improves human interpretation.

Pipeline Processing Stage	Mean Latency (ms)	Proportion (%)
Preprocessing & Normalization	0.35 ms	1.50%
Dual Representation Extraction	3.12 ms	13.33%
Quality-Risk Assessment	2.45 ms	10.47%
Patch Saliency Localization	8.84 ms	37.78%
Evidence Faiss Retrieval	4.22 ms	18.03%
Evidence Aggregation & Ranking	4.42 ms	18.89%
Total Serial End-to-End Latency	23.40 ms	100.0%
P95 Tail Latency	28.30 ms	—

TABLE IX: End-to-End Latency Profile. Under the declared benchmark environment, the evaluated pipeline required a mean of 23.40 ms per image, with a P95 latency of 28.30 ms.

VI. DISCUSSION & DUAL COMPOSITION

The evaluated results support a specialization-oriented dual-representation architecture in which the frozen representations serve complementary downstream roles. Adapting representations to acquisition invariance projects out high-frequency sensor noise, which attenuates sensitivity to subtle image perturbations. A modular dual-representation architecture resolves this tension by assigning representations to their strongest evaluated roles without learning unverified fusion weights.

VII. SCIENTIFIC LIMITATIONS & BOUNDARIES

1. Train, validation, and test partitions contain identical declared alloy classes; unseen-specimen generalization is not established.
2. Controlled synthetic perturbations do not encompass all real-world physical microscope defects.
3. Human expert validation of scientific interpretation was not performed.
4. Evidence availability is an operational metric and does not establish visual correctness.
5. Evidence retrieval establishes geometric similarity; it does not constitute causal or diagnostic explanation.
6. Saliency bounding envelopes are model-derived suspicious regions, not physically confirmed defect boundaries.
7. OOD screening does not constitute open-world anomaly discovery.
8. Acquisition robustness is bounded to evaluated HCCI SEM conditions.
9. Absent metadata fields can limit evidence interpretation.
10. Measured latencies depend on the benchmark execution environment.
11. Protocol-M and Protocol-U retrieval metrics are not directly interchangeable.
12. The dual representation is a deterministic composition, not a learned fusion model.
13. No clinical diagnosis or medical decision-making claim is made.

VIII. CONCLUSION

SCI-INTEL provides a reproducible, scientifically verified platform for acquisition-aware scientific image retrieval and quality-aware curation.

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